ModelIt!

Teacher's Guide

Why Is Antibiotic Resistance Getting Worse?

Modeling Natural Selection, Mutation, and the Evolution of Resistant Bacteria

Quick Reference

Grade Level	Grade 12L1
NGSS Standard	HS-LS4-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS4-2

Construct an explanation based on evidence that the process of evolution primarily results from four factors: (1) the potential for a species to increase in number, (2) the heritable genetic variation of individuals in a species due to mutation and sexual reproduction, (3) competition for limited resources, and (4) the proliferation of those organisms that are better able to survive and reproduce in the environment.

Three-Dimensional Learning

Science Practice	**Constructing Explanations** — Students construct evidence-based explanations for how the four factors of evolution — population growth potential, heritable variation, competition, and differential survival — drive antibiotic resistance.
Disciplinary Core Idea	**LS4.B Natural Selection** — Natural selection acts on heritable variation in bacterial populations: antibiotic exposure selects for resistance mutations, and surviving bacteria reproduce to increase resistance gene frequency in subsequent generations.
Crosscutting Concept	**Cause and Effect** — Students identify causal relationships between antibiotic usage patterns and the rate of resistance evolution, predicting how changes in human behavior affect bacterial population dynamics.

CAST Testing Alignment

MULTIPLE CHOICE:

A patient takes antibiotics for 5 of the prescribed 10 days and feels better. Based on the natural selection model, what is the most likely composition of the remaining bacterial population and why does this matter?

CONSTRUCTED RESPONSE:

Using your model, explain why giving low doses of antibiotics to healthy livestock to promote growth is considered one of the greatest threats to human health. Reference at least three model components and the mechanism of natural selection.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Antibiotics transformed medicine after Alexander Fleming's discovery of penicillin in 1928, turning previously fatal infections into treatable conditions. But bacteria have been evolving resistance mechanisms for billions of years — antibiotic resistance genes have been found in 30,000-year-old permafrost bacteria. What's changed is the speed of evolution: human antibiotic use has created unprecedented selective pressure that accelerates resistance development from millennia to mere years.

The Mechanics of Resistance

Bacteria develop resistance through several mechanisms: enzymatic degradation (producing enzymes like beta-lactamase that break down antibiotics), target modification (altering the cellular structure an antibiotic targets), efflux pumps (actively pumping antibiotics out of the cell), and reduced permeability (changing membrane proteins to prevent antibiotic entry). These mechanisms arise from random mutations during DNA replication and are passed to daughter cells during binary fission.

The Role of Horizontal Gene Transfer

Unlike humans, bacteria can share genetic material between unrelated species through three mechanisms: conjugation (direct cell-to-cell transfer via pili), transformation (uptake of free DNA from dead bacteria), and transduction (transfer via bacteriophage viruses). This means a resistance gene that evolves in a harmless soil bacterium can transfer to a deadly pathogen. Plasmids — small circular DNA molecules — often carry multiple resistance genes simultaneously, creating multi-drug resistance in a single transfer event.

Agricultural Antibiotics: The Hidden Driver

Approximately 73% of all antibiotics sold globally are used in livestock — not to treat sick animals, but as growth promoters at sub-therapeutic doses. This practice exposes trillions of bacteria to constant low-level selective pressure, the perfect conditions for gradual resistance evolution. Resistant bacteria then spread to humans through direct contact, food supply, water runoff, and airborne transmission. The WHO has called for a complete ban on growth-promotion antibiotics, but economic interests have slowed policy changes.

The Post-Antibiotic Threat

The WHO has warned that without urgent action, we face a post-antibiotic era where common infections and minor injuries can once again kill. The development pipeline for new antibiotics is nearly empty because pharmaceutical companies find it unprofitable — a new antibiotic takes 10-15 years and $1 billion to develop but can become obsolete in 2-3 years. Currently, 700,000 people die annually from resistant infections, projected to reach 10 million by 2050 — surpassing cancer as a cause of death.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify antibiotic exposure, bacterial population diversity, mutation rate, resistance gene frequency, and treatment effectiveness as the key components of the antibiotic resistance system.

E **Establish Relationships**

Students determine that antibiotic exposure creates selective pressure favoring resistant mutations, that incomplete courses and low doses accelerate resistance, and that treatment effectiveness decreases as resistance gene frequency rises.

V **Visualize & Model**

Students build a computational model showing how antibiotic usage patterns interact with bacterial reproduction and mutation to drive the evolution of resistance over multiple generations.

E **Evaluate Outcomes**

Students run full treatment, incomplete course, and agricultural overuse scenarios to test predictions about which behaviors drive resistance evolution fastest.

R **Revise & Extend**

Students add horizontal gene transfer, immune response, or biofilm formation to explore how real-world complexity affects resistance dynamics and treatment outcomes.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Every year, 1.27 million people die from infections that antibiotics can no longer treat. Today we investigate why."*

Watch For

Show cover image. Ask: Has anyone ever stopped taking antibiotics early? Why?

Slide 2: Learning Objectives

3 min

Say: *"Here's what you'll be able to do by the end of today's lesson — model how bacteria evolve resistance in real time."*

Watch For

Have students read objectives aloud. Pre-teach vocabulary with visual aids showing bacterial structures.

Slide 3: Big Question

3 min

Say: *"Are we heading toward a world where a simple cut could kill you because no antibiotic works? Let's find out."*

Watch For

Quick-write: Students write their initial hypothesis about why resistance is increasing. Save to compare later.

Slide 4: LEVER Framework

2 min

Say: *"We're going to model evolution by natural selection — happening right now, in real time, inside your body."*

Watch For

Preview each LEVER step. Emphasize that this is evolution you can observe in days, not millions of years.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this evolutionary system? Which inputs come from outside the bacteria?"*

Watch For

Guide sorting of external vs. internal components. Use think-pair-share for reasoning about which factors bacteria control vs. which humans control.

Slide 6: Activity 2: Connections

8 min

Say: *"When Antibiotic Exposure increases but the course is incomplete, what happens to the surviving population's Resistance Gene Frequency?"*

Watch For

Students draw arrows showing +/- relationships. Discuss the feedback loop: more resistance leads to less effective treatment leads to more antibiotic use.

Slide 7: Activity 3: Simulation

10 min

Say: *"Now let's test our model. Run each scenario — full course, incomplete course, and farm antibiotics — and record what happens."*

Watch For

Students run all scenarios, record predictions vs. observations. Discuss why incomplete courses are so dangerous.

Slide 8: Discoveries

5 min

Say: *"Our model reveals that WE are the selective pressure. Every time we misuse antibiotics, we train bacteria to defeat them."*

Watch For

Class discussion of discoveries. Compare initial hypotheses to model evidence. Emphasize this is evolution by natural selection in action.

Slide 9: STEM Challenge

12 min

Say: *"Now apply what you've learned. Your county needs a campaign that changes the specific behaviors driving resistance."*

Watch For

Groups design evidence-based campaigns targeting the three main drivers. Present and evaluate using model data.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Antibiotic Exposure, Bacterial Population Diversity
- **INTERNAL:** Mutation Rate, Resistance Gene Frequency, Treatment Effectiveness

Relationships

- **Antibiotic Exposure to Resistance Gene Frequency:** POSITIVE (+) — Antibiotic exposure creates selective pressure that kills susceptible bacteria and allows resistant mutants to survive and reproduce, increasing the proportion of resistance genes in the population.
- **Bacterial Population Diversity to Mutation Rate:** POSITIVE (+) — More diverse populations have greater genetic variation, increasing the probability that resistance-conferring mutations already exist or will arise through recombination.
- **Mutation Rate to Resistance Gene Frequency:** POSITIVE (+) — Higher mutation rates increase the probability of resistance mutations arising in each generation, providing more raw material for natural selection to act upon.
- **Resistance Gene Frequency to Treatment Effectiveness:** NEGATIVE (-) — As the proportion of resistant bacteria increases, the antibiotic kills fewer and fewer organisms, reducing treatment effectiveness until it reaches zero for fully resistant populations.

Key Discoveries

- Incomplete antibiotic courses are one of the most powerful drivers of resistance — they kill weak bacteria and leave the most resistant ones to reproduce without competition
- Resistance evolution follows predictable natural selection: random mutation provides variation, antibiotics provide selective pressure, and differential survival increases resistance gene frequency
- Horizontal gene transfer allows resistance to jump between bacterial species, meaning resistance that evolves in harmless soil bacteria can transfer to deadly pathogens
- Sub-therapeutic antibiotic use in agriculture exposes trillions of bacteria to low-level selective pressure — the perfect training ground for resistance evolution on a massive scale

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Bacteria become resistant by adapting to antibiotics like building immunity"	Bacteria don't 'learn' or 'adapt' to antibiotics the way humans build immunity to diseases. Resistance arises from random mutations that happen to provide a survival advantage when antibiotics are present. Natural selection favors these pre-existing mutants — the antibiotic doesn't cause the mutation, it selects for it. This is a critical distinction between Lamarckian (acquired traits) and Darwinian (natural selection) evolution.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Antibiotic resistance only affects people who take antibiotics"	Resistant bacteria spread between people, animals, and environments regardless of individual antibiotic use. A person who has never taken an antibiotic can be infected by MRSA from a gym locker room, resistant E. coli from contaminated food, or resistant tuberculosis from a cough. Resistance is a community-level threat driven by total population antibiotic use, not an individual consequence.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We can always develop new antibiotics to stay ahead of resistance"	The antibiotic development pipeline is nearly empty. Only 12 new antibiotics have been approved since 2017, and most are modifications of existing classes rather than truly novel mechanisms. Development takes 10-15 years and costs over $1 billion, but a new antibiotic generates less revenue than drugs for chronic conditions because it's used short-term and doctors are told to use it sparingly. Meanwhile, bacteria can evolve resistance in 2-3 years. We cannot innovate our way out of this crisis without also reducing misuse.